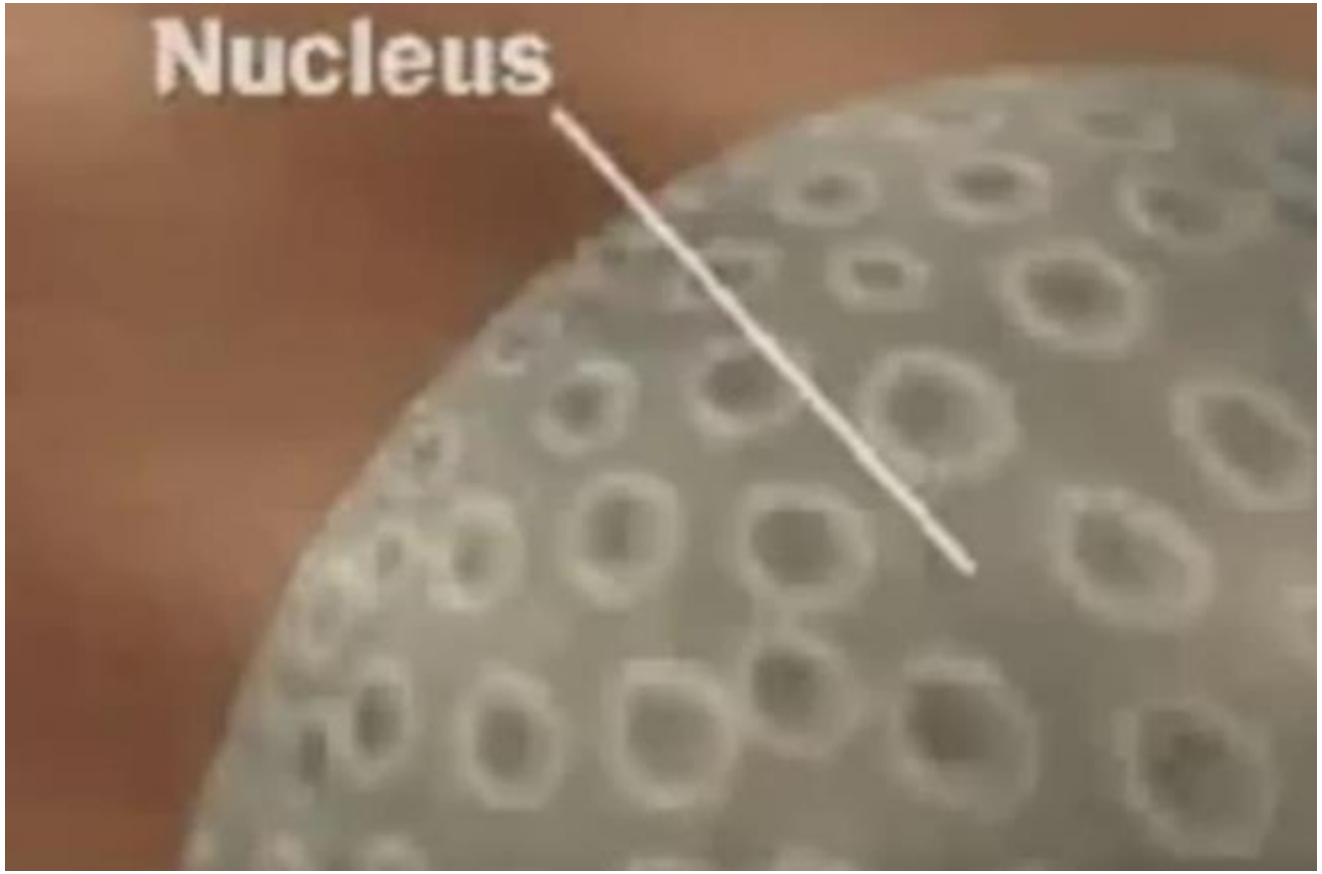
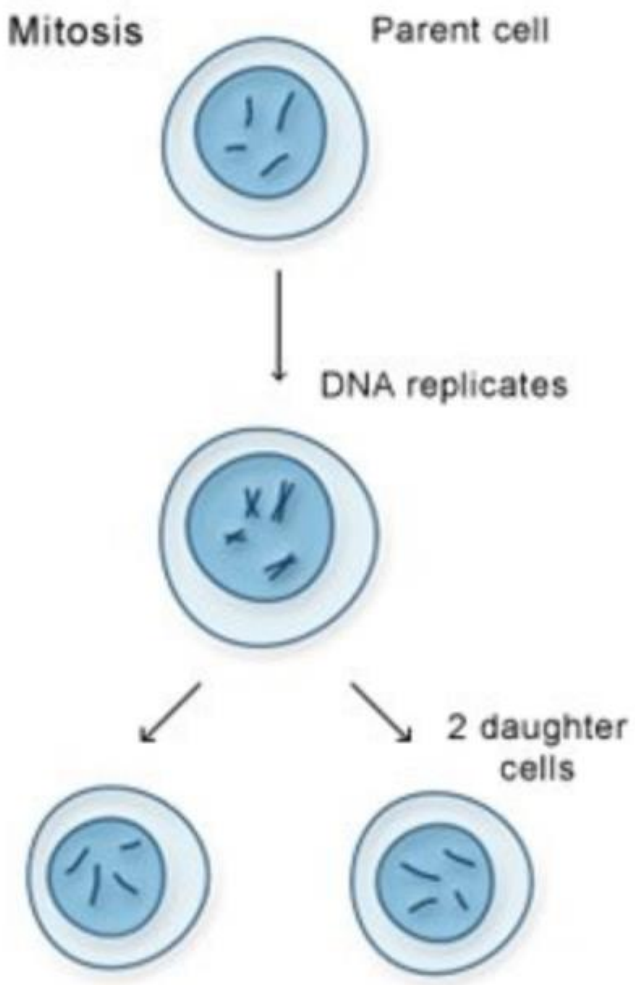

Nucleus

DNA Replication

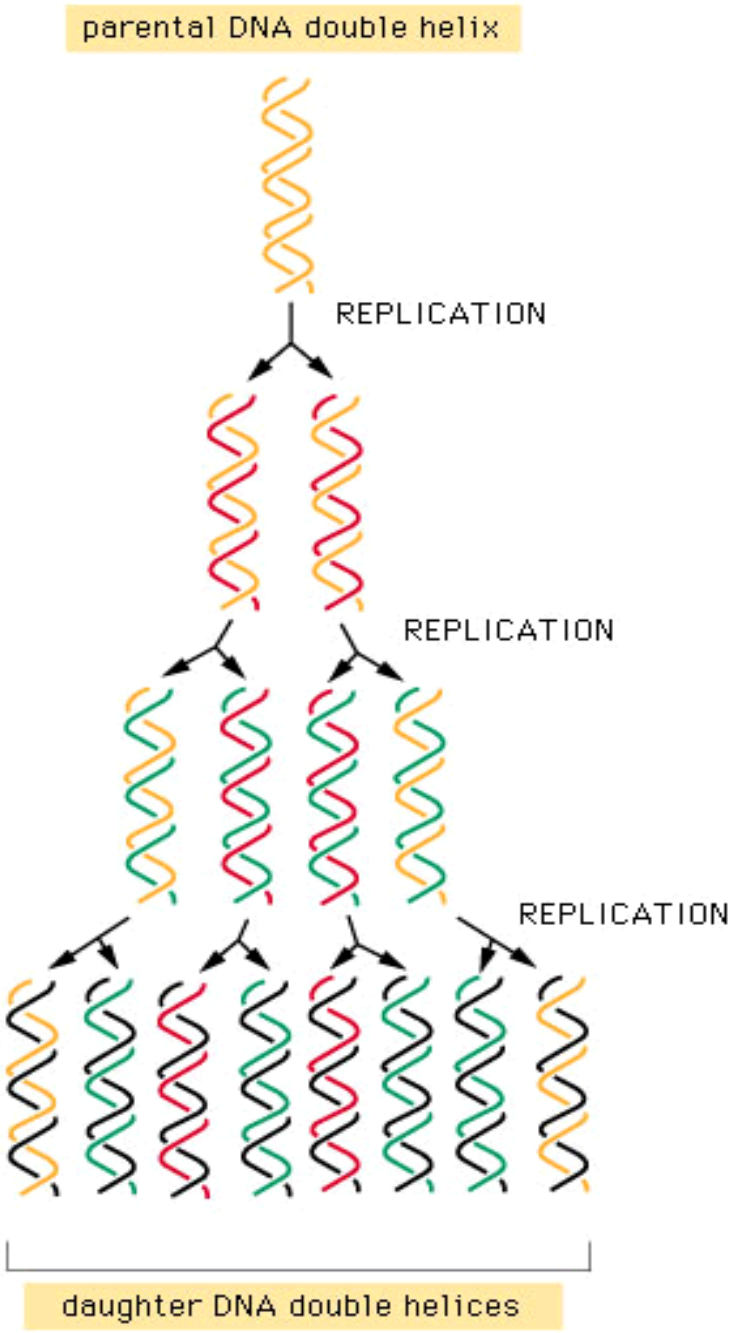
- Cell Division (mitosis)
  - ✓ Cells must copy their chromosomes (**DNA synthesis**) before they divide so that each daughter cell will have a copy
  - ✓ A region of the chromosome remains uncopied (**centromere**) in order to hold the **sister chromatids** together
    - – Keeps chromatids organized to help make sure each daughter cell gets exactly one copy
    - – **Nondisjunction** is when sister chromatids do not assort correctly and one cell ends up with both copies while the other cell ends up with none



U.S. National Library of Medicine

DNA Replication

- Semiconservative
- Daughter DNA is a double helix with 1 parent strand and 1 new strand
- Found that 1 strand serves as the template for new strand



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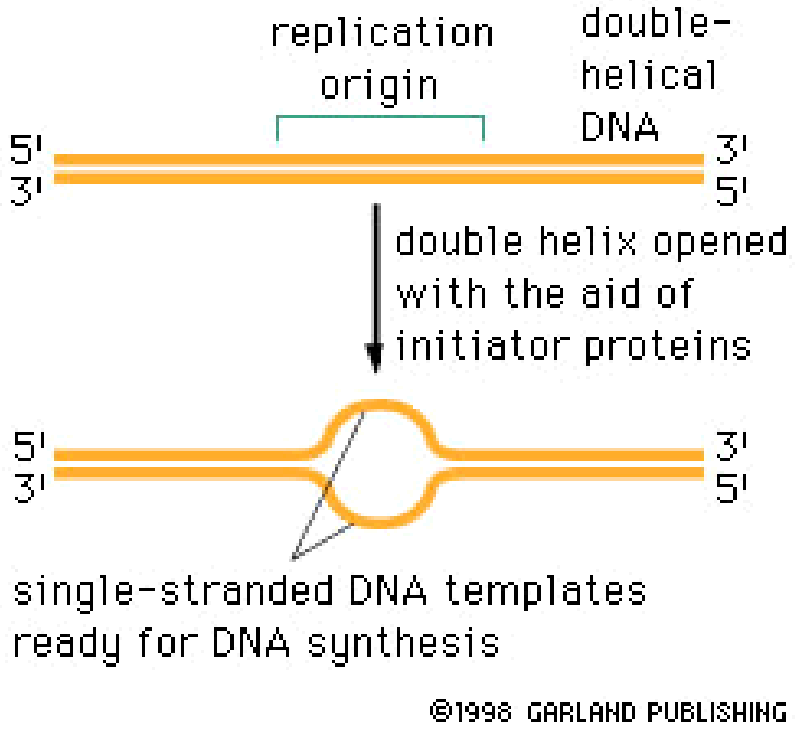
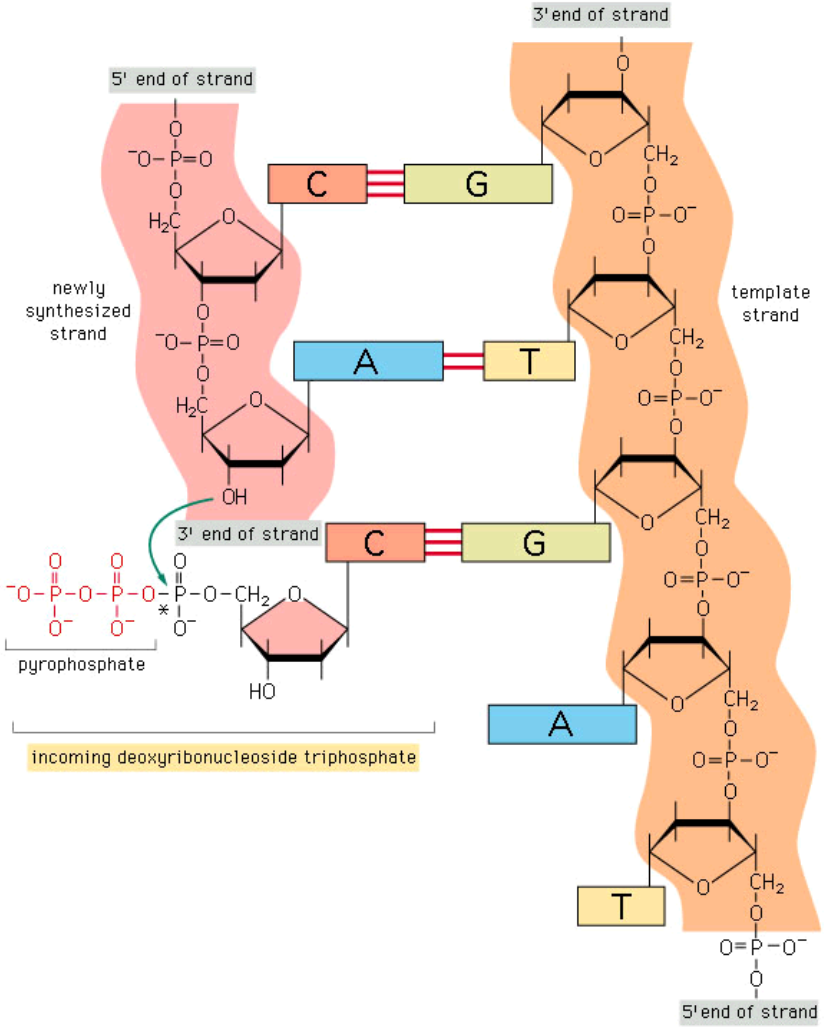
Replication Origin

- Site where replication begins
  - 1 in E. coli
  - 1,000s in human
- Strands are separated to allow replication machinery contact with the DNA
  - Many A-T base pairs because easier to break 2 H-bonds that 3 H-bonds
- Note anti-parallel chains



DNA Polymerase

- An enzyme that catalyzes the addition of a nucleotide to the growing DNA chain
- Nucleotide enters as a nucleotide tri-PO_4



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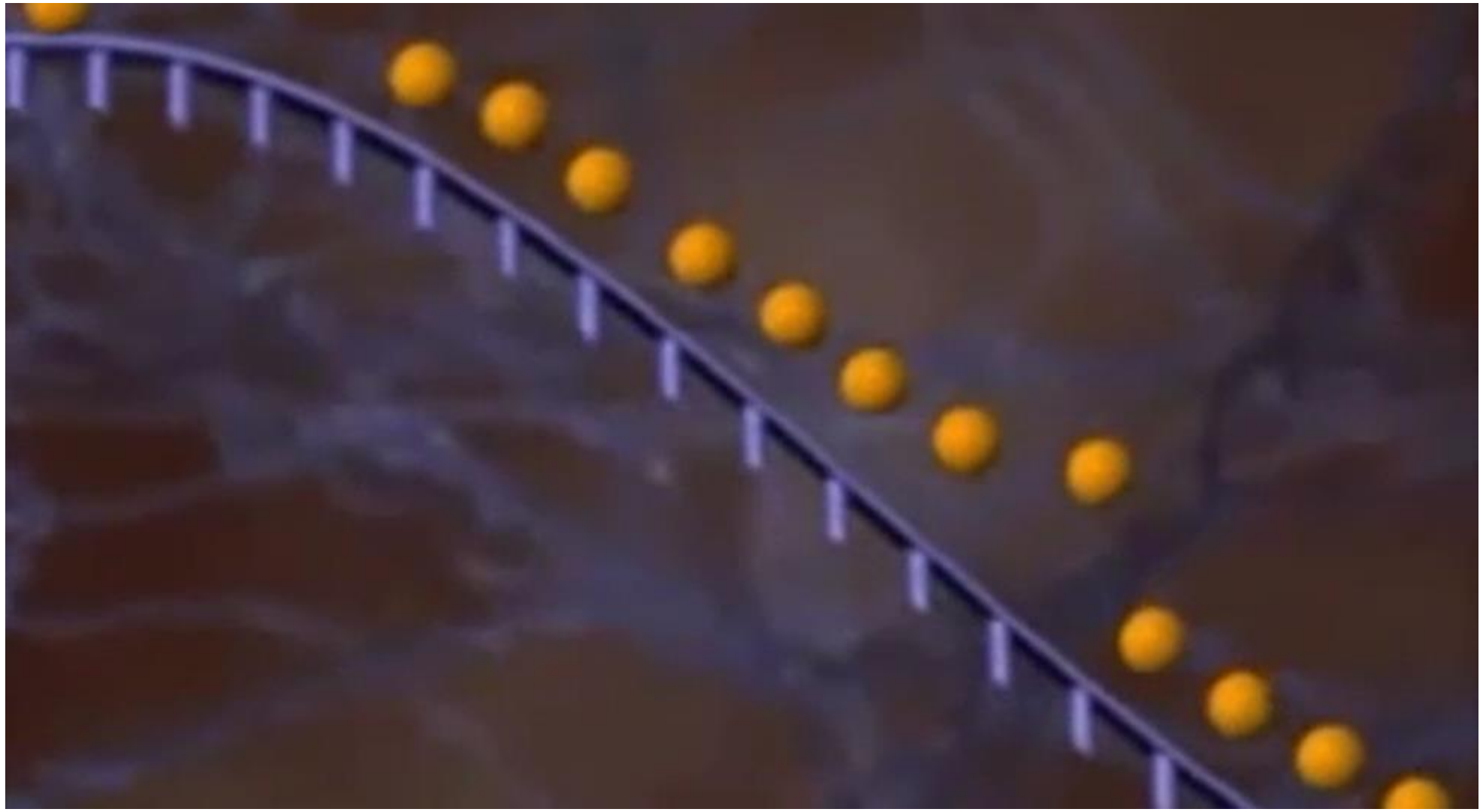
DNA Polymerase

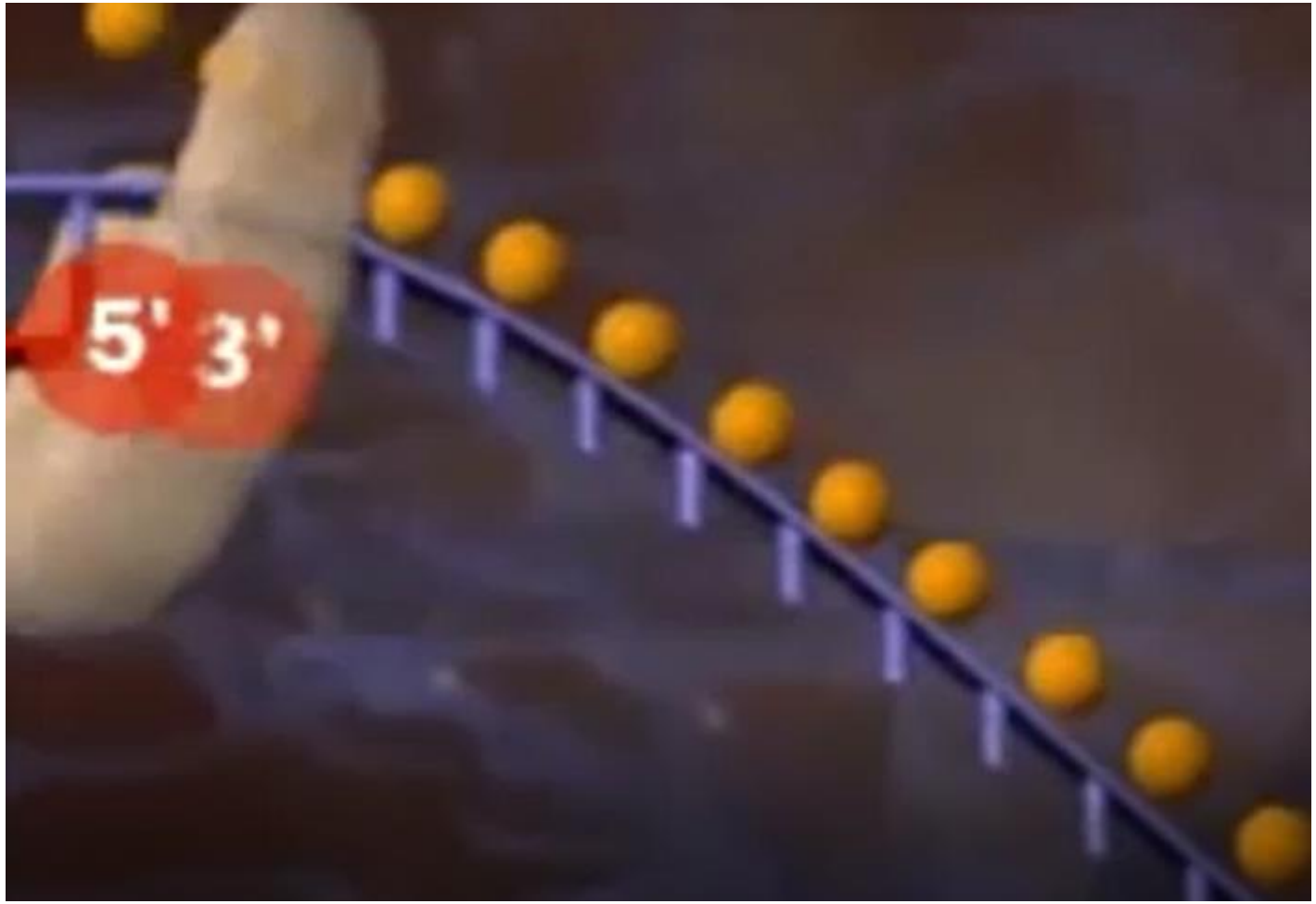
- Bidirectional synthesis of the DNA double helix
- Corrects mistaken base pairings
- Requires an established polymer (small RNA primer) before addition of more nucleotides
- Other proteins and enzymes necessary

HELICASE OPENING DOUBLE HELIX

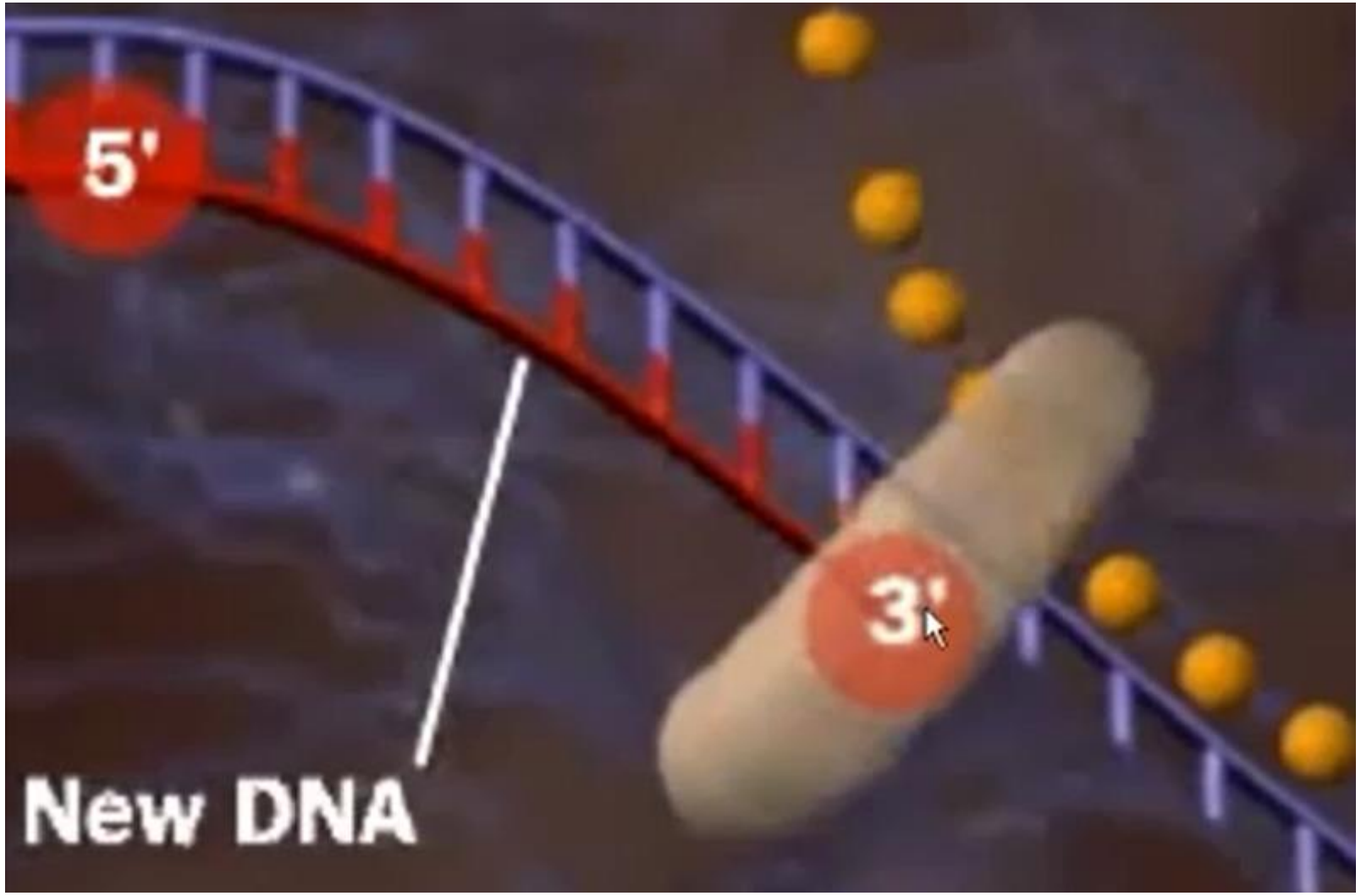
SINGLE STRAND BINDING PROTEINS

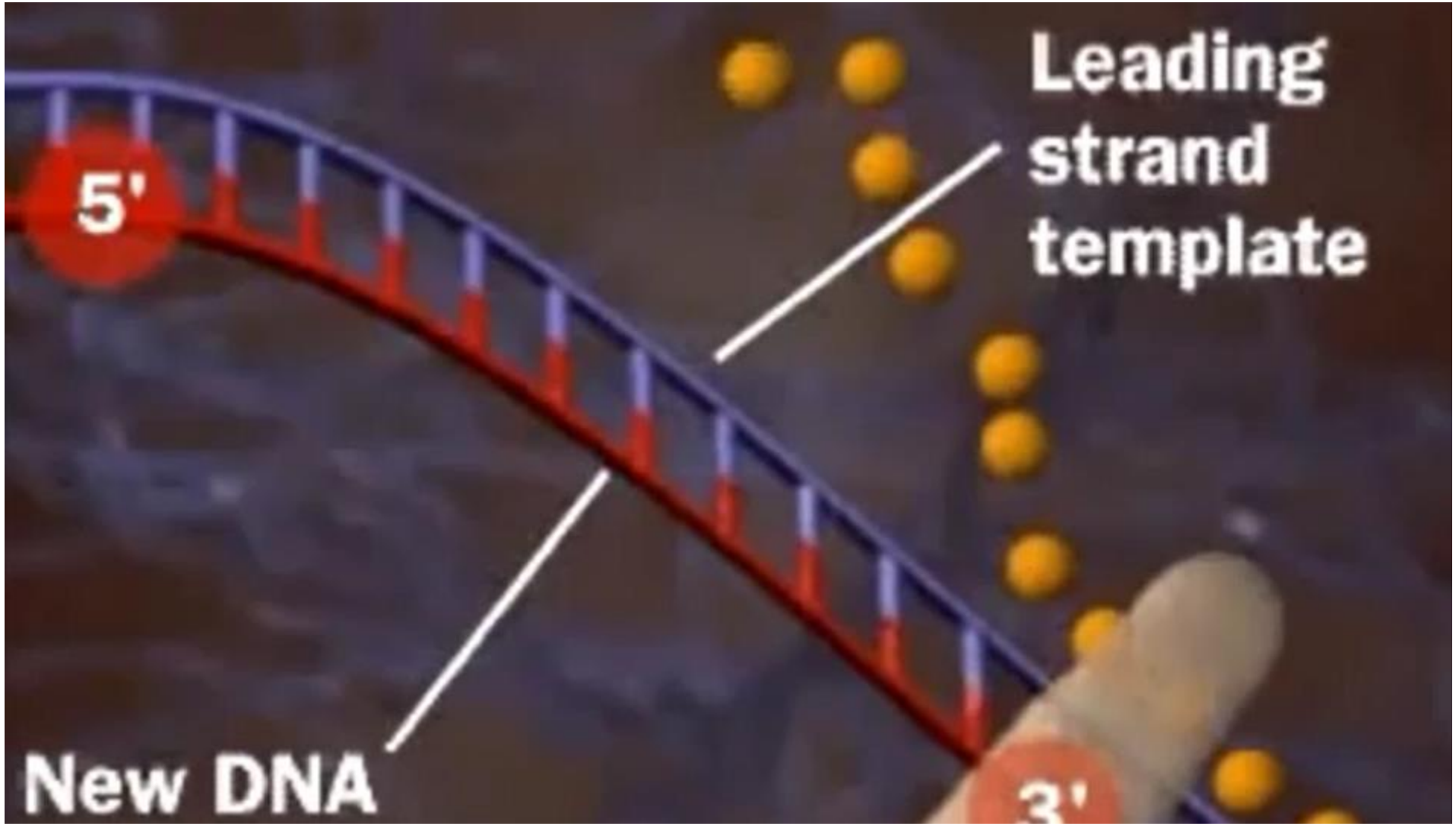


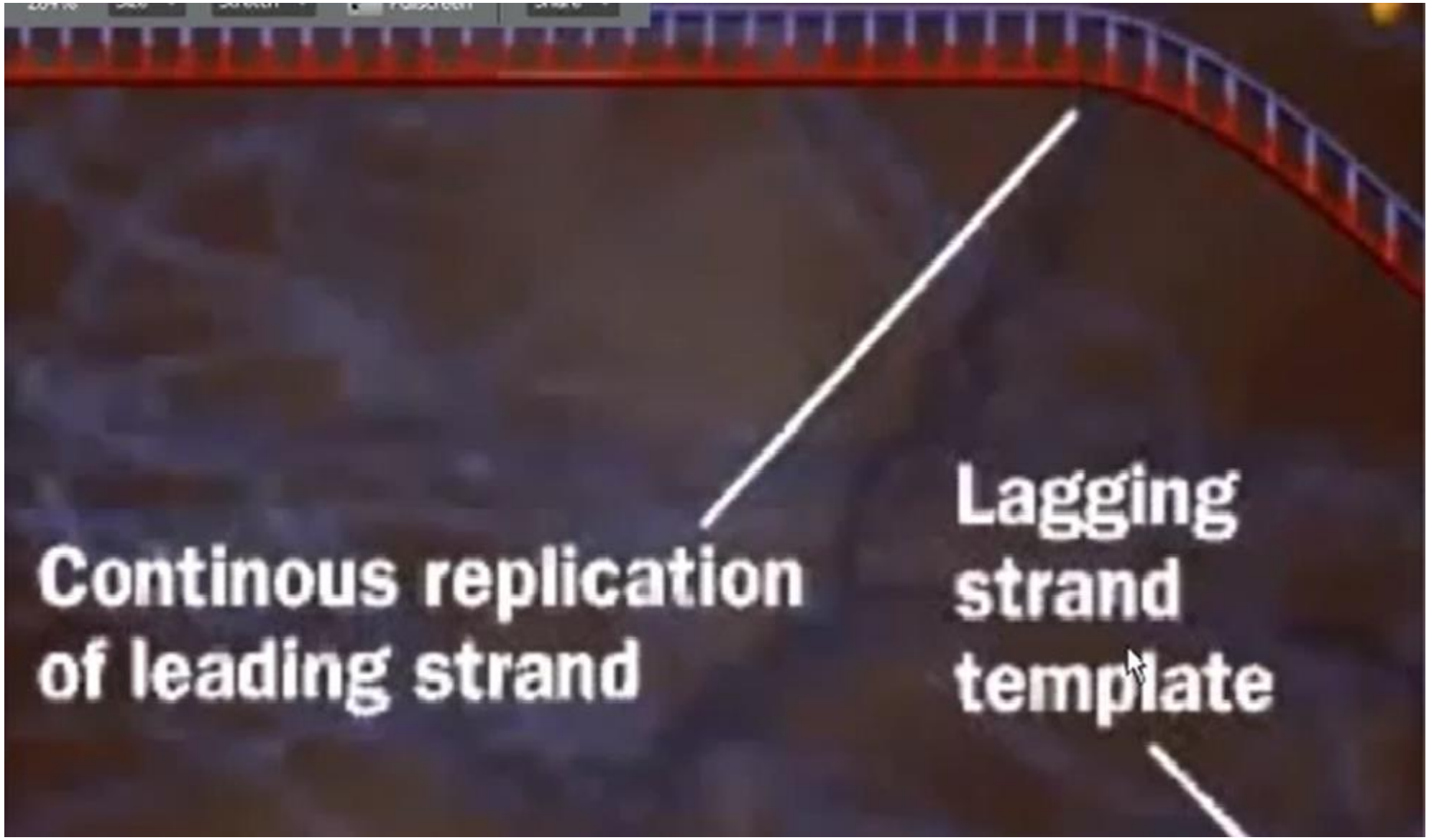
POLYMERASE BINDING TO SINGLE STRANDED DNA



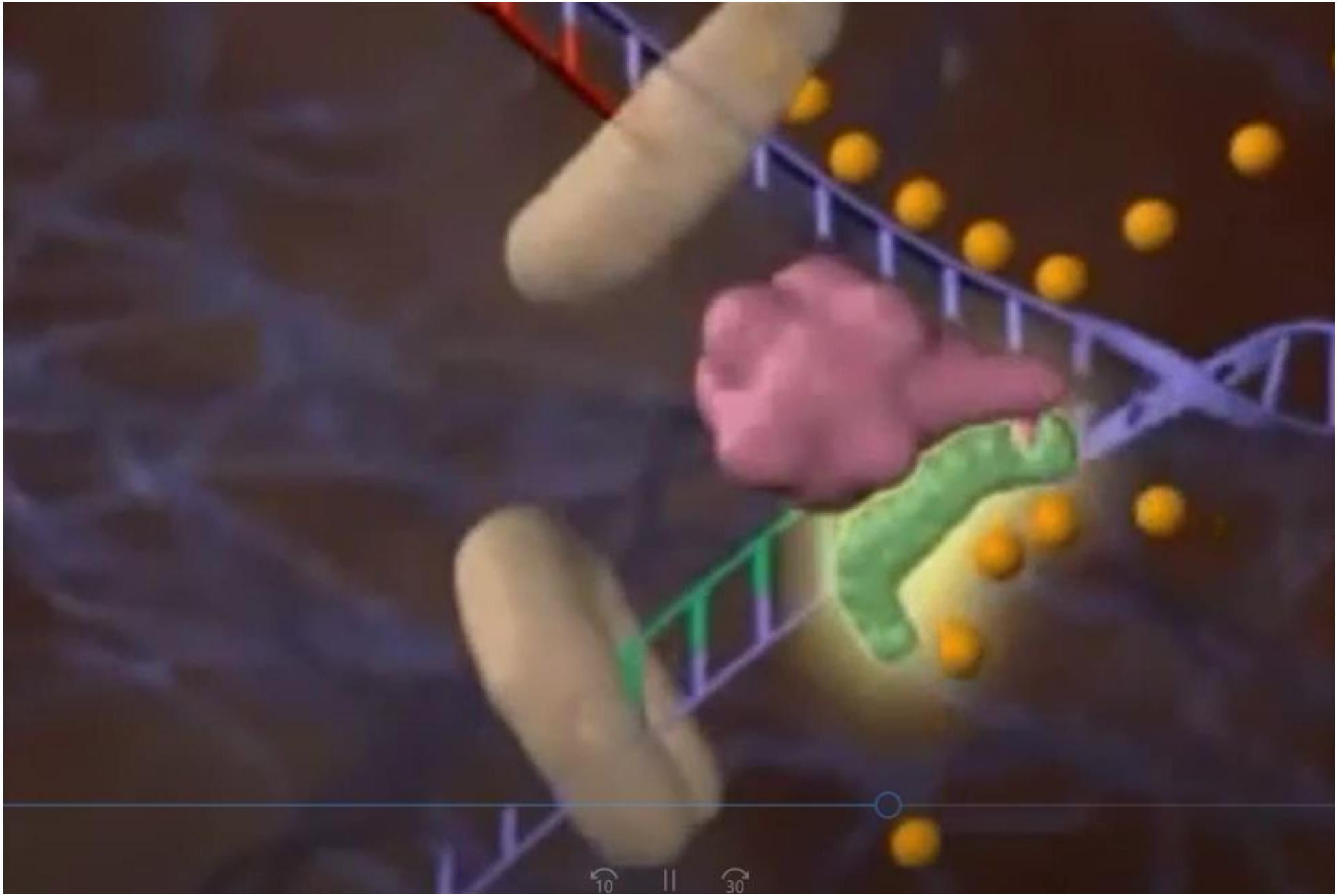
NEW DNA STRAND BEING FORMED

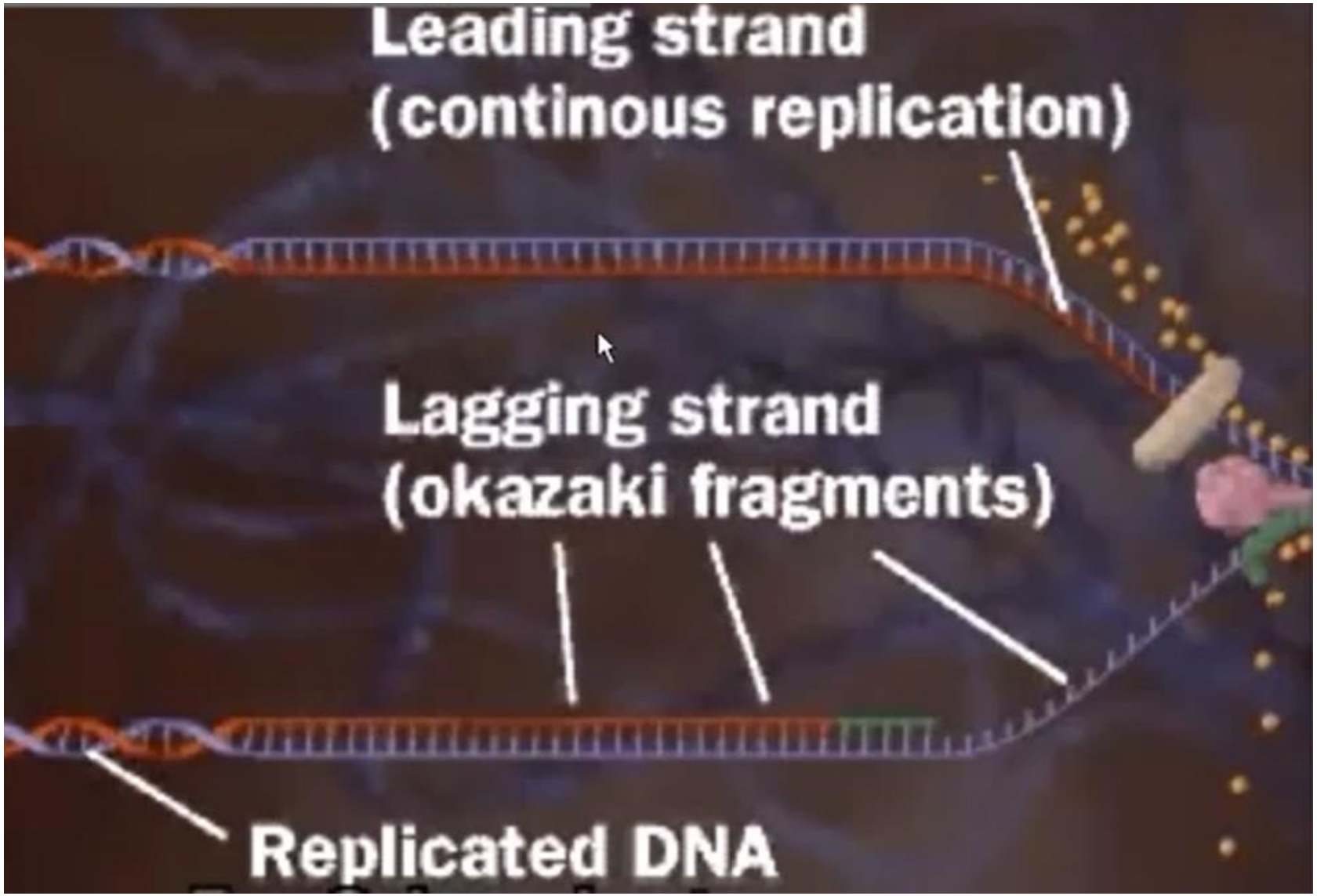



Leading strand template
5'
New DNA
3'

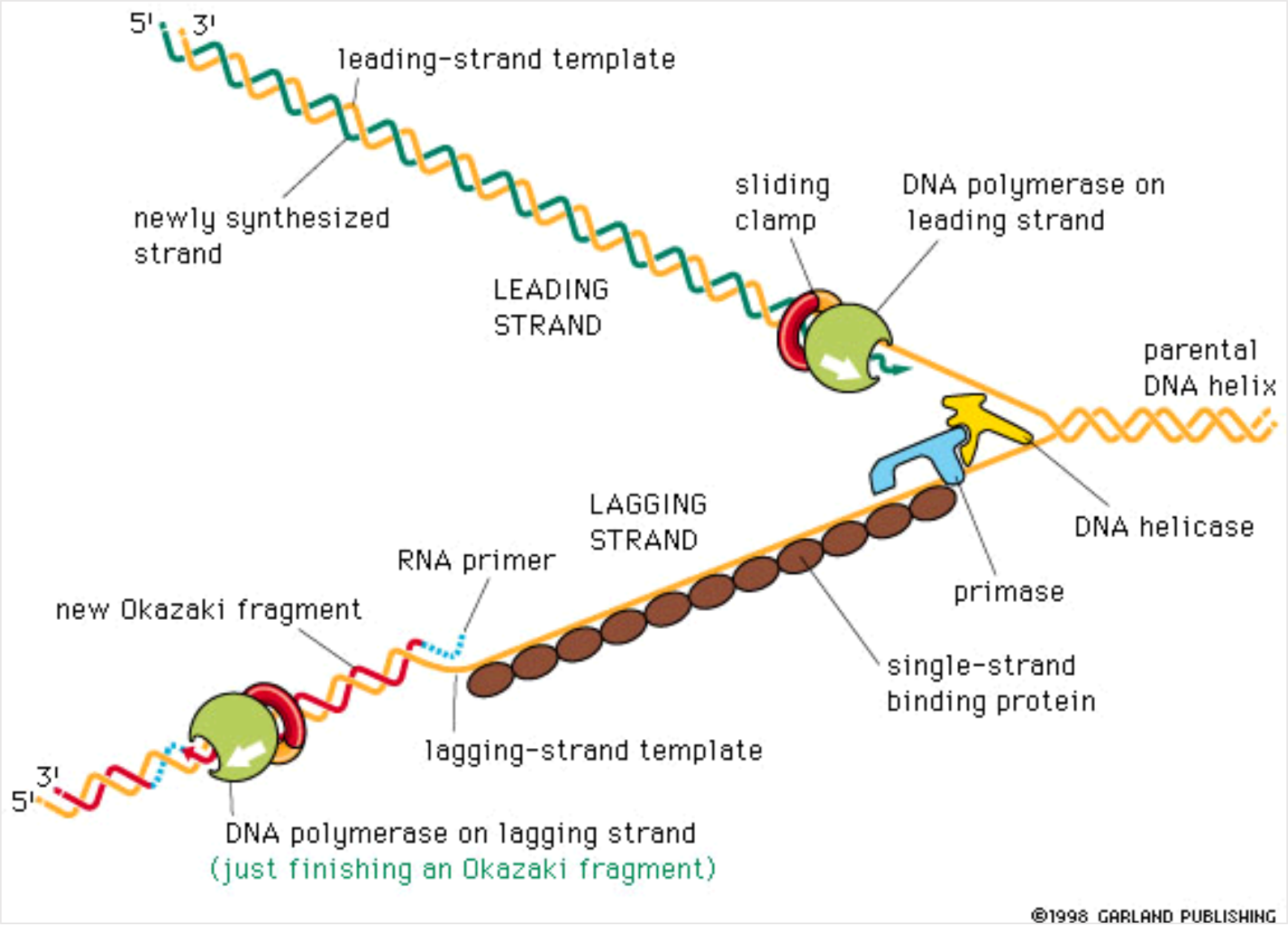

Continous replication
of leading strand
Lagging
strand
template

Okazaki
fragments




Leading strand
(continous replication)
Lagging strand
(okazaki fragments)
Replicated DNA

Components of the DNA Replication



DNA Errors

- Replication is not 100% accurate
  - Errors occur when copying DNA
- DNA damaged
  - Ultraviolet light
  - X-rays
  - Cigarette smoke
  - Nitrates (smoked meat)
  - Pesticides & Herbicides

Effects of Mutation

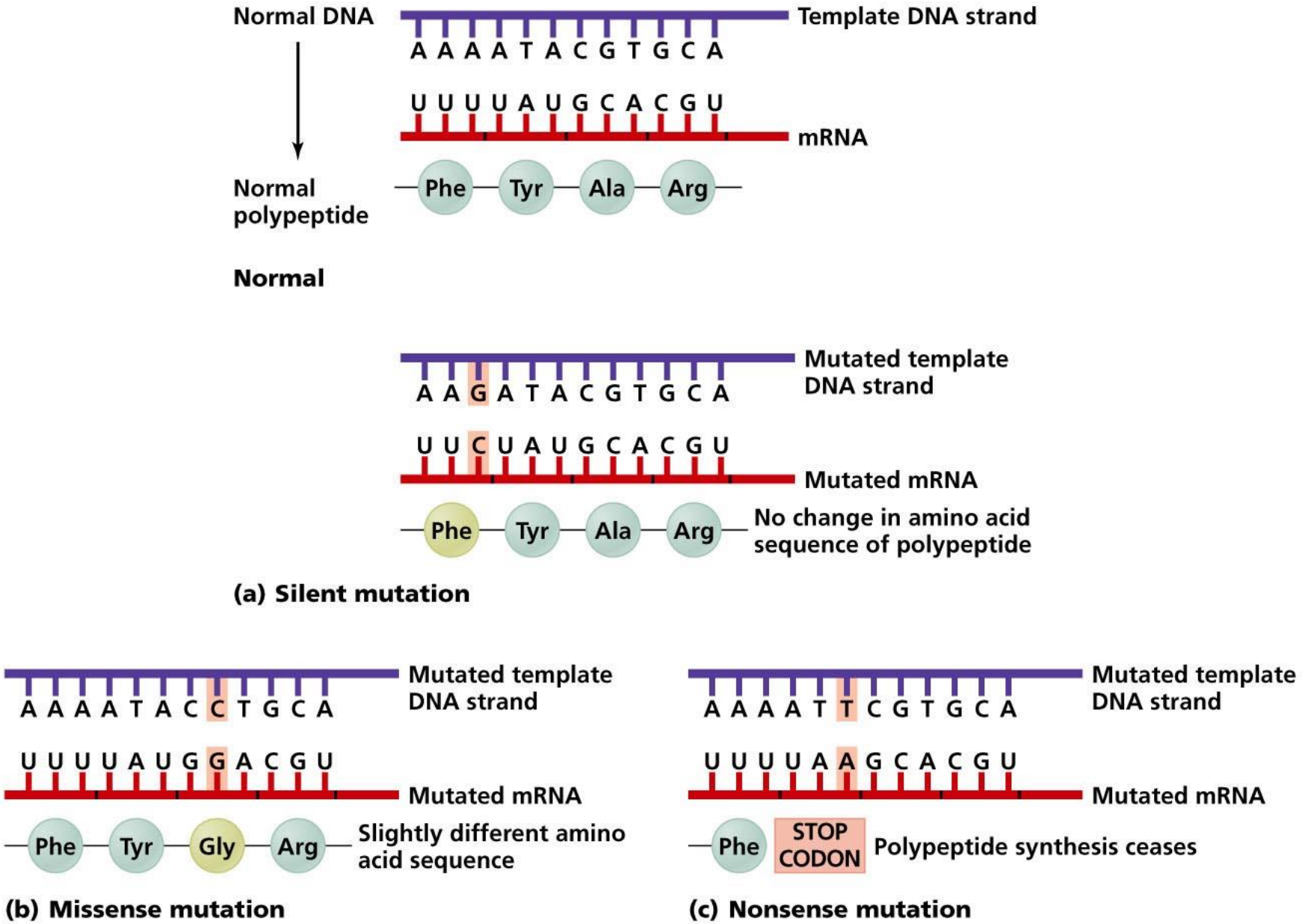
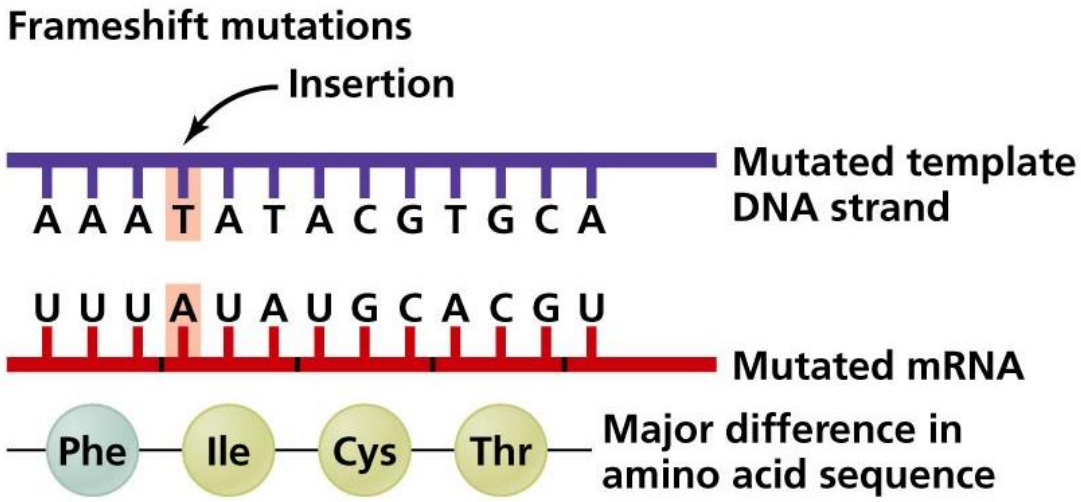
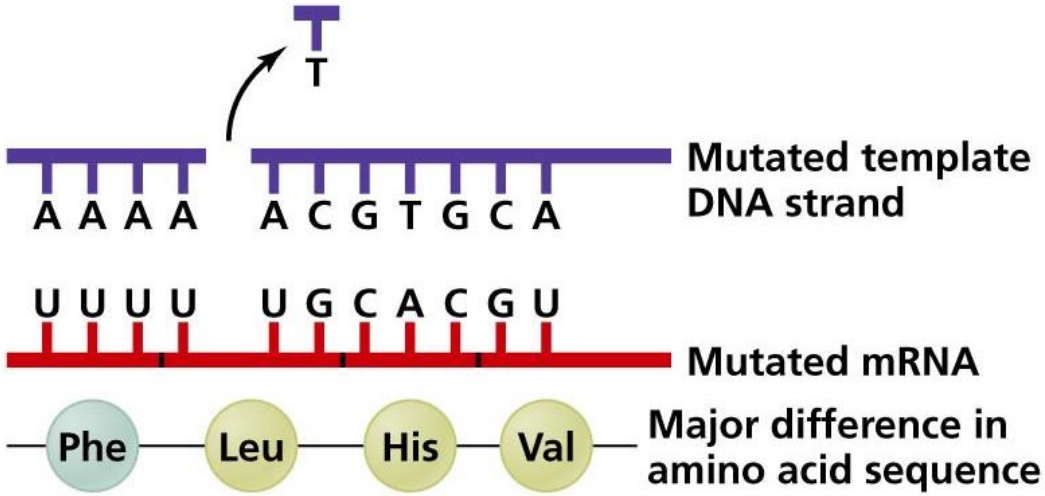


Figure 7.21a-c

Effects of Mutation



(d) Frameshift insertion



(e) Frameshift deletion

Figure 7.21d-e

An English Example

THEBOYANDTHEDOGRANFAR

THE BOY AND THE DOG RAN FAR

Missense:

THE BOY AND THE HOG RAN FAR

Nonsense:

THE BOY AND .

English Example-Frameshift

THEBOYANDTHEDOGRANFAR

THE BOY AND THE DOG RAN FAR

DEletion:

THE BOY ADT HED OGR ANF AR

Insertion:

THE EBO YAN DTH EDO GRA NFA R

Mutagens

- Radiation
  - Ionizing radiation – induces breaks in chromosomes
  - Nonionizing radiation – induces pyrimidine dimers
- Chemical mutagens
  - Nucleotide analogs – disrupt DNA and RNA replication and cause point mutations
  - Nucleotide-altering chemicals – result in base-pair substitution mutations and missense mutations
  - Frameshift mutagens – result in nonsense mutations

DNA Repair

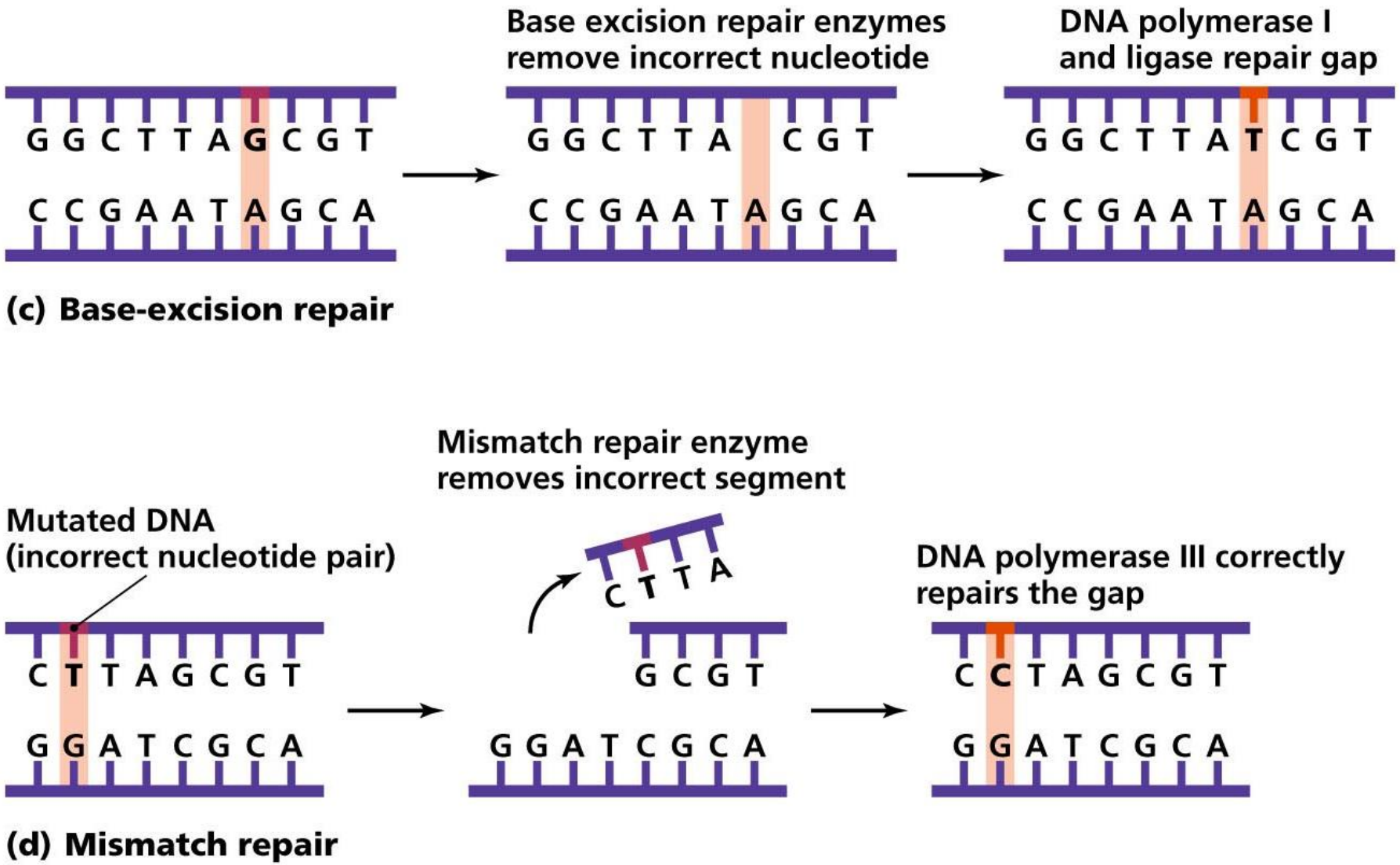


Figure 7.25c-d